

Assignment 4

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Problem (Munkres 23.10). Let $\{X_\alpha\}_{\alpha \in J}$ be an indexed family of connected spaces; let X be the product space

$$X = \prod_{\alpha \in J} X_\alpha.$$

Let $a = (a_\alpha)$ be a fixed point of X .

- (a) Given any finite subset K of J , let X_K denote the subspace of X consisting of all points $x = (x_\alpha)$ such that $x_\alpha = a_\alpha$ for $\alpha \notin K$. Show that X_K is connected.
- (b) Show that the union Y of the spaces X_K is connected.
- (c) Show that X equals the closure of Y . Conclude that X is connected.

Proof of (a). For $K \subseteq J$, finite, points in X_K coincide with a_α on all but finitely many α .

Let $f : X_K \rightarrow \{0, 1\}$ be any continuous function. We will work to show that f must be constant to conclude that X_K is connected.

Suppose there are two points $x = (x_\alpha)$, $y = (y_\alpha) \in X_K$ such that $f(x) \neq f(y)$.

We suppose, without loss of generality, that $f(x) = 0$ and $f(y) = 1$.

As $\{0\}$, $\{1\}$ are open in the codomain, the fibers of 0 and 1 are open sets in X_K who happen to contain x and y , respectively.

Then,

$$f^{-1}(0) = \prod_{\alpha} U_\alpha$$

where finitely many $U_\alpha \neq X_\alpha$. Similarly

$$f^{-1}(1) = \prod_{\alpha} U_\alpha$$

with finitely many $U_\alpha \neq X_\alpha$.

Note: This is way to complicated to be correct, find another way. $\square$